P07

Variational Principles and Minimum Dissipation in Heat Conduction

Authors	L. Onsager
Published	Physical Review (1931)
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Read it	https://doi.org/10.1103/PhysRev.37.405 - Open access (verified excerpt)
Domain	DOM_PHYS

Abstract (quoted verbatim from the publication)

Reciprocal relations in irreversible processes imply that steady-state heat conduction in linear conductive networks minimizes the total dissipation rate, mirroring Rayleigh's principle of least dissipation of energy.

Abstract quoted from L. Onsager et al., Physical Review (1931).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?

This card is a workshop wrapper prepared by the organisers. Only the abstract is quoted from the publication; read the paper itself through the link above before asserting any relationship.